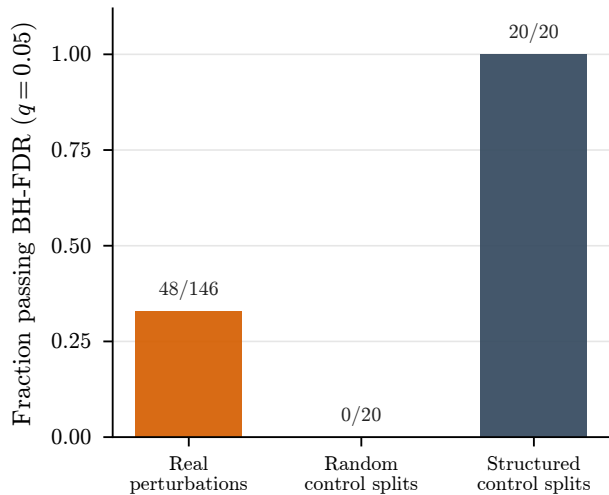
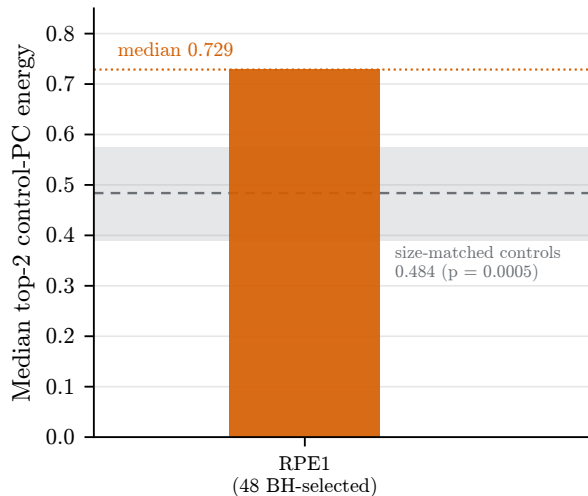


(a) All structured control splits
pass BH-FDR



(b) Selected shifts concentrate in
leading control PCs



Random splits pass 0/20; structured splits pass 20/20, as expected for PC-tail subsets. Alignment beyond size-matched controls is an association, not proven causation.